

Application of Incremental Learning to Medical Image Segmentation of Glioblastoma Multiforme

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Abstract — In this work, we investigated the impact of Incremental Learning on improving the precision of GBM image segmentation. We began by using three distinct models—U-Net, SegResNet, and Swin-Unet—within the MONAI framework, training them on publicly available datasets. These models were further trained using the CGMH-GBM hospital dataset to enhance their segmentation accuracy. After implementing incremental learning, we noticed a significant improvement in the U-Net model's performance when it was combined with the BraTS 2021 dataset and data from Chang Gung Memorial Hospital. On the other hand, the Swin-Unet model showed improvement in some cases but also displayed potential signs of overfitting, especially when used with the UPenn-GBM and hospital datasets. Notably, a marked decrease in performance was observed when the Swin-Unet model was applied to the combined UCSF-PDGM dataset, which includes tumor images ranging from Grade II to IV. This suggests that the Swin-Unet model might struggle to adapt to the complexity of tumor grades in the UCSF-PDGM dataset. It also highlights the importance of incremental learning in enhancing the segmentation capabilities of specific models on specific datasets.

I. INTRODUCTION

Glioblastoma Multiforme (GBM), a tumor occurring in the adult central nervous system, is notorious for its high aggression and deadly nature [1]. It can be divided into three main morphological areas: Whole Tumor (WT), Enhancing Tumor (ET), and Tumor Core (TC) [4].

This represents one of the significant challenges in contemporary diagnostic radiology. As medical imaging techniques advance, including Magnetic Resonance Imaging (MRI) and Computed Tomography (CT), the medical community continually strives to discover more precise methods for early diagnosis and treatment of brain tumors.

Deep learning models have achieved advances in medical image segmentation in recent years, but real-world applications are limited due to the scarcity of GBM-specific data. The manual annotation process by radiologists requires much time and effort and is susceptible to subjectivity and biases [13].

In our work, we utilize the cutting-edge open-source medical framework Medical Open Network for AI (MONAI) for GBM medical image segmentation [11]. We compared specific models based on CNN-based U-Net, highlighting its outstanding performance, which led to more precise GBM imaging segmentation. We also incorporated data from Chang Gung Memorial Hospital for incremental learning techniques, resulting

in the continuous improvement of the model's performance.

This work provides a background on brain tumors and GBM, followed by an overview of the existing radiomic methods. Following that, we delve into using incremental learning in medical image segmentation, providing a detailed explanation of our methodology and experimental design. To summarize, we present our results and highlight promising directions for future work.

II. THE PROPOSED APPROACH

2.1 Incremental Learning to Medical Image Segmentation of Glioblastoma Multiforme

As shown in Fig. 1, our work commenced with acquiring image data from public datasets and initiating training with medical image AI models. Subsequently, we enhanced the model's segmentation accuracy by incorporating Keelung Chang Gung Memorial Hospital glioblastoma (CGMH-GBM) data through incremental learning, culminating in the final segmentation output.

The objective of medical image segmentation is to divide the image into meaningful segments, identifying critical structures or regions. Precise segmentation is crucial for accurate diagnosis and treatment strategy of GBM. Moreover, MRI sequences—including native (T1), T2-weighted (T2), post-contrast T1-weighted (T1C), and Fluid Attenuated Inversion Recovery (FLAIR) — offer unique insights into the tumor and surrounding tissues, which are instrumental in pinpointing the tumor core and devising surgical strategies [4].

In fact, radiologists manually label these images, which is a process that is not only labor-intensive but also prone to biases. Despite the revolutionary impact of deep learning on medical image segmentation, the scarcity of specialized GBM data continues to be a challenge, limiting the broader implementation of these techniques [3].

The following sections will explore three distinguished models in this field. U-Net is characterized by its U-shaped architecture, comprising an encoder for feature extraction and a decoder for creating segmentation masks; its feature connections preserve essential contextual information, bolstering segmentation accuracy [5]. SegResNet, a ResNet-based variant, employs deep residual blocks to enhance feature extraction while maintaining spatial information. It produces high-resolution feature maps that retain image details, making it well-suited for multi-class segmentation tasks [6].

Moving on to Swin UNETR, a neural network grounded in the Transformer architecture, it merges the U-Net structure with the self-attention mechanism of Transformers, facilitating parallel processing and multi-scale feature extraction for superior performance [7].

After these models were verified to replicate the findings of principal studies, our focus shifted to parameter optimization and fine-tuning, specifically targeting GBM MRI scans. This rigorous calibration hastened the training phase for the GBM dataset, markedly enhancing the precision of segmentation.

To tackle these challenges, we have integrated incremental learning into our methodology. This technique enables models to acclimate to new data without discarding previously learned information gradually. This is particularly advantageous in cases like GBM, where annotated data is scarce, offering a strategic edge. As additional data becomes available and integrated, the model is progressively fortified, boosting segmentation accuracy.

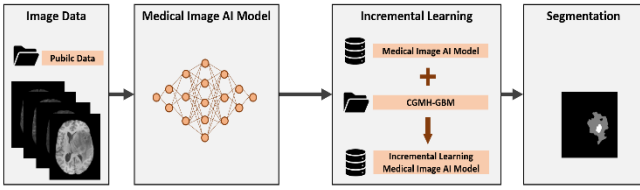


Fig. 1 Brain tumor segmentation architecture in incremental learning

2.2 Optimization via Incremental Learning

In analyzing GBM medical images, which are complex and variable, traditional one-time learning models often need more data and better generalization capabilities. Compared to these conventional methods, our introduced incremental learning strategy demonstrates its unique advantages in real-world hospital settings where data is scarce. In fact, Traditional learning approaches are frequently impractical in medical imaging due to their reliance on extensively annotated datasets, which are often inadequate and costly to obtain.

In comparison, the incremental learning approach permits the model to start with a concentrated focus on a limited dataset. This approach equips the model with the capability to evolve and enhance its recognition of new features as additional data is gradually incorporated [11]. This progressive expansion of the learning scope doesn't merely improve the model's precision in identifying specific GBM image features but significantly bolsters its adaptability to various tumor imaging types.

In Figure 2, it provides a step-by-step overview of the incremental learning approach utilizing the U-Net architecture. For this study's experimental setup, we initially trained the U-Net model using the **Brain Tumor Segmentation Challenge 2021** (BraTS 2021) [4] dataset to ensure that the model acquired a foundational understanding of tumor characteristics and segmentation capabilities.

Next, we employed the CGMH-GBM hospital dataset in the incremental learning phase to further refine the model. This strategic approach not only enabled our model to achieve high accuracy on the initial BraTS 2021 dataset but also exhibited exceptional adaptability and room for improvement when introduced to new hospital data with varying characteristics.

This incremental learning approach has enabled our model to adeptly overcome typical challenges in medical image analysis, thereby enhancing the precision of GBM image segmentation and the generalization ability of the model.

These findings underscore the transformative potential of incremental learning within medical imaging, paving the way for models that can more adeptly handle the nuances and variability inherent to clinical data. Our method showcases the advantages of a carefully fine-tuned learning approach, which can gradually integrate new data while preserving and building upon previously acquired knowledge.

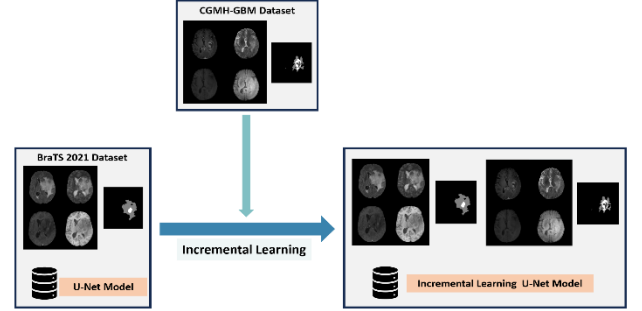


Fig. 2 Incremental Learning based on the U-Net Model

2.3 Metrics Evaluation

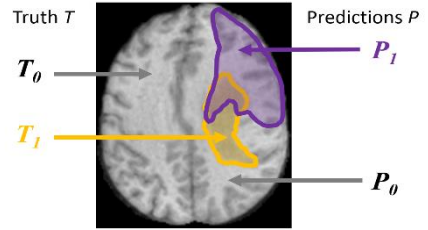


Fig. 3 Metrics for T_1 lesion & P_1 prediction. Overlap: $P_1 \cap T_1$

Performance Metrics:

For each of the three tumor regions, we generated a binary map from the algorithmic predictions, $\mathcal{P} \in \{0,1\}$, and the consensus ground truth among experts, $\mathcal{T} \in \{0,1\}$. Based on this, we evaluated the well-established Dice score [3].

$$Dice(\mathcal{P}, \mathcal{T}) = \frac{|\mathcal{P}_0 \cap \mathcal{T}_1|}{|\mathcal{P}_1| + |\mathcal{T}_1| / 2} \quad (1)$$

In this content, $\wedge$ signifies the logical AND operation. The notation $|\cdot|$ denotes the cardinality of a set, which translates to the count of voxels within it. $\mathcal{P}_1$ and $\mathcal{T}_1$ are the voxel sets where $\mathcal{P}$ and $\mathcal{T}$ have values of 1, respectively (Fig. 3).

The Dice score adjusts the count of true positives relative to the mean size of both segmented regions. This metric mirrors the F score—the harmonic means of the precision-recall curve—and can seamlessly convert to the Jaccard index.

We further computed sensitivity, often termed the actual positive rate, and specificity, commonly called the valid negative rate.

$$Sens(\mathcal{P}, \mathcal{T}) = \frac{|\mathcal{P}_1 \cap \mathcal{T}_1|}{|\mathcal{T}_1|} \wedge Spec(\mathcal{P}, \mathcal{T}) = \frac{|\mathcal{P}_0 \cap \mathcal{T}_0|}{|\mathcal{T}_0|} \quad (2)$$

where and represent voxels where $\mathcal{P} = 0$ and $\mathcal{T} = 0$, respectively.

The Dice coefficient, true positive rate, and true negative rate are metrics for assessing the congruence of voxels within segmented areas. Another category of metrics gauges the gap between the edges of segmentation, that is, the boundary distance.

An illustrative instance is the Hausdorff metric, which determines for every point p on the boundary $\partial\mathcal{P}_1$ of a specified volume $\mathcal{P}_1$ the nearest least-squares distance $d(p, t)$ to points t on the boundary $\partial\mathcal{T}_1$ of another specified volume $\mathcal{T}_1$, and the reverse, ultimately yielding the supreme value across all distances d

$$Haus(\mathcal{P}, \mathcal{T}) = \max \left\{ \sup_{p \in \partial\mathcal{P}_1} \inf_{t \in \partial\mathcal{T}_1} d(p, t), \sup_{t \in \partial\mathcal{T}_1} \inf_{p \in \partial\mathcal{P}_1} d(t, p) \right\} \quad (3)$$

However, computing the apex of all boundary distances renders the Hausdorff metric highly sensitive to minor, isolated subareas within either $\mathcal{P}_1$ or $\mathcal{T}_1$. For instance, in our analysis of the 'active tumor' zone, $\mathcal{P}_1$ and $\mathcal{T}_1$ could be composed of several diminutive regions or irregular shapes with a significant boundary-to-volume ratio. In the assessment of the 'entire tumor,' forecasts containing scant erroneous positive areas—those that have a negligible impact on the segmentation's aggregate quality since they can be eliminated through suitable postprocessing—may still exert a significant effect on the cumulative Hausdorff figure. Consequently, we adopted a more resilient variant of the Hausdorff metric—indicating not the extreme boundary distance between $\mathcal{P}_1$ and $\mathcal{T}_1$, but rather the 95th percentile of it.

As depicted in Fig. 3, it is used to compute the Dice score, sensitivity and specificity. The $\mathcal{T}_1$ region represents the true lesion, highlighted with an orange outline, while $\mathcal{T}_0$ denotes the remaining normal area. $\mathcal{P}_1$ signifies the area predicted as the lesion, outlined in red, and $\mathcal{P}_0$ is identified as usual. On the right lateral side of the lesion, $\mathcal{P}_0$ and $\mathcal{P}_1$ overlap, corresponding to the area labeled as $\mathcal{P}_1 \cap \mathcal{T}_1$ in the definition of the Dice score.

III. EXPERIMENTS AND RESULTS

3.1 Implement Details and Dataset

Datasets:

In our work, we employed a combination of both public and private datasets. Specifically, from the public datasets, we used a total of 2,357 patients, which includes 1,251 patients from BraTS 2021, 611 patients from The University of Pennsylvania glioblastoma (UPenn-GBM) [9], and 501 patients from The University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) [8]. Additionally, from our proprietary dataset, CGMH-GBM, we used 82 patients. It's important to note that while UCSF-PDGM includes tumor grades II to IV, the other datasets primarily consist of Grade IV tumors.

Data Pre-processing:

In this work, all images from the CGMH-GBM dataset were subjected to a meticulously crafted series of preprocessing steps before model training to ensure consistency, accuracy, and efficiency in the analysis. As presented in Fig. 4, we adopted the

BraTS preprocessing pipeline within the Cancer Imaging Phenomics Toolkit (CaPTk) [12].

Initially, through N4 Bias Field Correction, we rectified brightness inhomogeneities induced by the scanner; this was promptly followed by the implementation of Z-Score Normalization to harmonize the intensity scales across different scanning devices. Then, Registration was performed to precisely align the scans from various patients in the spatial dimension and to standardize them to a uniform resolution of $240 \times 240 \times 155$ (240x240 represents the size of the images, while 155 indicates the number of slices in each NIFTI file); after that, the Skull-Stripping process was conducted to remove non-brain tissues from the images, preserving only the crucial brain structures.

Finally, Label Binarization was employed to transform tumor region labels into a binary format. The comprehensive preprocessing protocol not only heightened the learning efficiency of the subsequent machine-learning model but also set a clear benchmark for high-precision tumor segmentation.

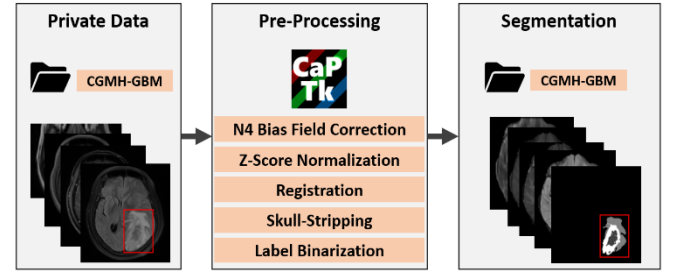


Fig. 4 Using the CaPTk tool in Medical Image Processing

Experimental Tools & Environment:

Our models were predominantly trained under the frameworks of MONAI 1.2.0 and PyTorch 1.12.1. On the hardware front, we leveraged a single NVIDIA RTX A5000 24GB, along with eight NVIDIA Tesla V100 16GB, as our computational resources.

3.2 Results

In this work, we conducted a preliminary assessment of the performance of three principal deep learning models: U-Net, SegResNet, and Swin-Unet, in the task of brain tumor segmentation. Before implementing incremental learning, these models were evaluated on three different datasets: BraTS 2021, UPenn-GBM, and UCSF-PDGM. This evaluation assessed their accuracy in segmenting the TC, WT, and ET, as presented in Table I.

TABLE I
THE PERCENTAGE OF MODELS

Model	Average Dice	Dice TC	Dice WT	Dice ET
U-Net	84.29% ¹	82.49% ¹	89.64% ¹	80.76% ¹
	87.79% ²	84.87% ²	92.53% ²	85.98% ²
	78.19% ³	74.16% ³	86.40% ³	74.93% ³
SegResNet	71.95% ¹	69.75% ¹	77.83% ¹	69.86% ¹
	80.03% ²	77.48% ²	84.54% ²	79.08% ²
	69.51% ³	66.82% ³	76.10% ³	67.55% ³
Swin-Unet	79.15% ¹	74.32% ¹	84.55% ¹	79.58% ¹
	74.38% ²	69.97% ²	79.68% ²	73.50% ²
	85.75% ³	79.80% ³	91.09% ³	87.13% ³

¹. BraTS 2021, ² UPenn-GBM, ³ UCSF-PDGM

Model Performance Comparison:

The U-Net model exhibited outstanding performance on the UPenn-GBM dataset, achieving an average Dice similarity coefficient of 87.79%, particularly impressive in segmenting the WT with a score of 92.53%. These results underscore U-Net's robust capability in delineating tumor-related regions. Additionally, the U-Net model attained an accuracy of 84.87% for the TC and 85.98% for the ET.

Conversely, the SegResNet model showed a relatively lower average Dice score of 80.03% on the UPenn-GBM dataset, with an accuracy of 84.54% for WT segmentation and 78.09% for ET segmentation. This suggests that the SegResNet model may require further adjustments and optimization for certain segmentation tasks.

Meanwhile, the Swin-Unet model reached an average Dice score of 85.75% on the UCSF-PDGM dataset, accomplishing 91.09% accuracy for WT segmentation and 87.13% for ET segmentation.

The results after adding the Chang Gung Memorial Hospital dataset are shown in Table II.

TABLE II
THE PERCENTAGE OF MODELS AFTER THE INCREMENTAL LEARNING

Model	Average Dice	Dice TC	Dice WT	Dice ET
U-Net	84.35% ¹	82.72% ¹	89.88% ¹	80.46% ¹
	87.59% ²	85.09% ²	92.21% ²	85.45% ²
	76.83% ³	72.95% ³	84.90% ³	72.84% ³
SegResNet	72.67% ¹	70.20% ¹	78.03% ¹	70.30% ¹
	78.32% ²	75.94% ²	82.13% ²	76.89% ²
	70.68% ³	67.48% ³	77.12% ³	67.81% ³
Swin-Unet	79.41% ¹	75.08% ¹	84.70% ¹	79.76% ¹
	85.03% ²	78.81% ²	90.58% ²	86.01% ²
	72.16% ³	67.71% ³	77.67% ³	71.10% ³

¹ BraTS 2021 + CGMH-GBM

² UPenn-GBM + CGMH-GBM

³ UCSF-PDGM + CGMH-GBM

Comparative Analysis Before and After Incremental Learning:

After implementing incremental learning, in the case of the U-Net model, there was an increase in the overall Dice score when combining BraTS 2021 data with hospital data. Although the overall Dice score slightly decreased when combining UPenn-GBM dataset with hospital data, it remained at a respectable level. In contrast, when UCSF-PDGM data was combined with hospital data, there was a substantial decrease in the overall Dice score.

In the SegResNet model, the overall Dice score increased when combining BraTS 2021 with hospital data, as well as when UCSF-PDGM was combined with hospital data. Conversely, a slight decrease in the overall Dice score was observed when UPenn-GBM data was combined with hospital data.

Within the Swin-Unet model, combining BraTS 2021 with hospital data resulted in a modest yet consistent improvement in the overall Dice score. Notably, the integration of UPenn-GBM with hospital data led to the most substantial enhancement in the average Dice score among all datasets, although there is a concern for potential overfitting.

Conversely, the integration of UCSF-PDGM with hospital data

led to a significant reduction in the overall Dice score. This reduction is associated with the unique attributes of the UCSF-PDGM dataset, which will be further elaborated in the discussion section.

3.3 Discussion

In the process of incremental learning, the utilization of the CGMH-GBM dataset markedly enhanced the performance of most of our models. Based on the comprehensive analysis above, we have outlined three directions for discussion:

U-Net Model Outperforms Other Models:

In our work, these results indicate that the U-Net model surpasses other models in the task of medical image segmentation of glioblastoma multiforme. This phenomenon is attributed to the U-Net model architecture, specifically designed for medical image segmentation, featuring a unique symmetric convolutional network structure that is more effective in capturing detailed features within images. Moreover, the U-Net model employs extensive skip connections, better preserving the contextual information in images, which is crucial for segmenting complex tumor structures. We will further analyze the performance advantages of U-Net across various aspects, including parameter efficiency, training time, and sensitivity to a range of image features.

The Importance of Incremental Learning in Glioblastoma Multiforme Image Segmentation:

Although incremental learning has been widely applied in medical image segmentation, most studies have not specifically focused on glioblastoma multiforme images. In our research, we have particularly explored how incremental learning can enhance the identification capabilities of this highly heterogeneous tumor. Images of glioblastoma multiforme exhibit irregular morphology and complexly varying boundaries, necessitating a more detailed learning process for models to adapt to these characteristics. Therefore, we believe that incremental learning plays a crucial role in this issue. It enhances the model's generalization ability and accuracy by gradually learning from the constantly changing data distribution.

Specificity of the CGMH-GBM Dataset and Analysis of Model Performance Variations:

The CGMH-GBM dataset, originating from Chang Gung Memorial Hospital, may differ from publicly available datasets such as BraTS 2021, UPenn-GBM, and UCSF-PDGM regarding imaging equipment settings, patient volumes, and tumor image processing methods. These variations can significantly impact the learning process of the model and the final effectiveness of segmentation. Our work aims to demonstrate that, despite facing data with high heterogeneity and diverse origins, incremental learning combined with these models can still achieve commendable segmentation results. However, from the comparison between Table I and Table II, we observed that after incremental learning, the performance of some models improved while others decreased. This is particularly evident in the UCSF-PDGM dataset, which contains tumor images ranging from Grade II to IV, whereas other datasets consist of Grade IV GBM. These varying levels of tumor features place higher demands on the recognition of the model and classification capabilities. Additionally, differences in architecture and parameter settings between models could also be factors contributing to these performance variations.

Alternatively, some aspects of the model's parameters or structure might not adequately respond to the nuances in the CGMH-GBM dataset. This divergence underscores the notion that different models might require distinct strategies or fine-tuning during training and optimization.

Additionally, it's essential to be aware of the risk of overfitting that can accompany incremental learning. Although the CGMH-GBM dataset enhanced the performance of many models, caution is needed to avoid excessive dependence on this dataset. To balance improved model performance and the risk of overfitting, it is advisable to employ strategies like incorporating a wider variety of datasets or focusing on model compression techniques.

In essence, while the CGMH-GBM dataset proved instrumental in elevating the performance of most models, it also shed light on some limitations inherent to specific models. These insights pave the way for future research and model optimization endeavors, emphasizing the importance of judiciously selecting and amalgamating datasets.

IV. CONCLUSION AND FUTURE WORK

In this study, we employ incremental learning to enhance the accuracy of segmenting glioblastoma multiforme (GBM) across both the CGMH-GBM and various public datasets. The results indicated a notable enhancement in the U-Net model's performance across numerous datasets. However, the Swin-Unet model generates mixed results. It demonstrated improvements under certain conditions, but exhibited significant performance declines in different scenarios, notably with the UCSF-PDGM dataset that includes tumor images from Grades II to IV. This variation in performance highlights the model's limited adaptability to different tumor grades. Additionally, there is a concern about the Swin-Unet model's tendency to overfit, which could explain the varied performance with the UCSF dataset.

Moving forward, we plan to refine the design and parameters of these models to boost their versatility and effectiveness across a wide range of datasets. Furthermore, we are considering the use of domain adaption methods to address the over-fitting problem.

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